

EXCEPT Operator in SQL Server with Examples

In this article, I am going to discuss the **EXCEPT Operator in SQL Server** with Examples. The EXCEPT operator belongs to the SET Operator category. Please read our previous article where we discussed the basics of [SET Operators in SQL Server](#) as well as we also discussed the [UNION and UNION ALL SET Operators](#) in detail.

What is the EXCEPT operator in SQL Server?

The EXCEPT operator in SQL Server is used to return unique rows from the left query which isn't present in the right query's results. If this is not clear at the moment then don't worry we will try to understand this with some examples.

Understanding the EXCEPT operator with examples.

Let us understand the EXCEPT Operator with an example. We are going to use the following "EmployeeIndia" and "EmployeeUK" tables to understand this operator.

EmployeeIndia				EmployeeUK			
ID	Name	Gender	Department	ID	Name	Gender	Department
1	Pranaya	Male	IT	1	James	Male	IT
2	Priyanka	Female	IT	2	Priyanka	Female	IT
3	Preety	Female	HR	3	Sara	Female	HR
4	Subrat	Male	HR	4	Subrat	Male	HR
5	Anurag	Male	IT	5	Pam	Female	HR

Please use the below SQL Script to create Database EmployeeDB, tables EmployeeIndia and EmployeeUK and populate these two tables with the required test data.

```
CREATE DATABASE EmployeeDB
GO
USE EmployeeDB
GO
CREATE TABLE EmployeeIndia
(
    ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Gender VARCHAR(10),
    Department VARCHAR(50)
)
GO
INSERT INTO EmployeeIndia VALUES(1, 'Pranaya', 'Male', 'IT')
INSERT INTO EmployeeIndia VALUES(2, 'Priyanka', 'Female', 'IT')
INSERT INTO EmployeeIndia VALUES(3, 'Preety', 'Female', 'HR')
INSERT INTO EmployeeIndia VALUES(4, 'Subrat', 'Male', 'HR')
```

```

INSERT INTO EmployeeIndia VALUES(5, 'Anurag', 'Male','IT')
GO
CREATE TABLE EmployeeUK
(
    ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Gender VARCHAR(10),
    Department VARCHAR(50)
)
GO
INSERT INTO EmployeeUK VALUES(1, 'James', 'Male','IT')
INSERT INTO EmployeeUK VALUES(2, 'Priyanka', 'Female','IT')
INSERT INTO EmployeeUK VALUES(3, 'Sara', 'Female','HR')
INSERT INTO EmployeeUK VALUES(4, 'Subrat', 'Male','HR')
INSERT INTO EmployeeUK VALUES(5, 'Pam', 'Female','HR')
GO

```

Example:

The following SQL Query will return the unique rows from the left query (the select statement before the EXCEPT operator) that is not present in the right query (the select statement after the EXCEPT operator).

```

SELECT ID, Name, Gender, Department FROM EmployeeIndia
EXCEPT
SELECT ID, Name, Gender, Department FROM EmployeeUK

```

Results:

ID	Name	Gender	Department
1	Pranaya	Male	IT
3	Preety	Female	HR
5	Anurag	Male	IT

To retrieve all of the rows from Table EmployeeUK that do not exist in Table EmployeeIndia, reverse the two queries as shown below.

```

SELECT ID, Name, Gender, Department FROM EmployeeUK
EXCEPT
SELECT ID, Name, Gender, Department FROM EmployeeIndia

```

Result:

ID	Name	Gender	Department
1	James	Male	IT
3	Sara	Female	HR
5	Pam	Female	HR

EXCEPT Operator in SQL Server on a single table:

You can also use the EXCEPT operator on a single table. Let's use the following Employees table for this example.

Id	Name	Gender	Salary
1	Mark	Male	52000
2	Mary	Female	55000
3	Steve	Male	45000
4	John	Male	40000
5	Sara	Female	48000
6	Pam	Female	60000
7	Tom	Male	58000
8	George	Male	65000
9	Tina	Female	67000
10	Ben	Male	80000

Please use the below SQL Script to create and populate the Employees table with the required test data.

```
-- SQL script to create Employees table
Create table Employees
(
    Id int identity primary key,
    Name nvarchar(100),
    Gender nvarchar(10),
    Salary int
)
Go
Insert into Employees values ('Mark', 'Male', 52000)
Insert into Employees values ('Mary', 'Female', 55000)
Insert into Employees values ('Steve', 'Male', 45000)
Insert into Employees values ('John', 'Male', 40000)
Insert into Employees values ('Sara', 'Female', 48000)
Insert into Employees values ('Pam', 'Female', 60000)
Insert into Employees values ('Tom', 'Male', 58000)
Insert into Employees values ('George', 'Male', 65000)
Insert into Employees values ('Tina', 'Female', 67000)
Insert into Employees values ('Ben', 'Male', 80000)
Go
```

Order By clause should be used only once after the last select statement

```
Select Id, Name, Gender, Salary
From Employees
Where Salary >= 50000
Except
Select Id, Name, Gender, Salary
From Employees
Where Salary >= 60000
order By Name
```

Result:

Id	Name	Gender	Salary
1	Mark	Male	52000
2	Mary	Female	55000
7	Tom	Male	58000

Difference between EXCEPT and NOT IN Operator SQL server

Now, you may have one question on your mind i.e. what is the difference between EXCEPT and NOT IN Operator in SQL Server. Let's understand the difference between EXCEPT and NOT IN operators in SQL Server with one example. We are going to use the following two tables (TableA and TableB).

TableA				TableB			
ID	Name	Gender	Department	ID	Name	Gender	Department
1	Pranaya	Male	IT	2	Priyanka	Female	IT
2	Priyanka	Female	IT	3	Preety	Female	HR
3	Preety	Female	HR				

Please use the below SQL Script to create the necessary tables with the required test data.

```
CREATE TABLE TableA
(
    ID INT,
    Name VARCHAR(50),
    Gender VARCHAR(10),
    Department VARCHAR(50)
)
GO
INSERT INTO TableA VALUES(1, 'Pranaya', 'Male','IT')
INSERT INTO TableA VALUES(2, 'Priyanka', 'Female','IT')
INSERT INTO TableA VALUES(3, 'Preety', 'Female','HR')
GO
CREATE TABLE TableB
(
    ID INT,
    Name VARCHAR(50),
    Gender VARCHAR(10),
    Department VARCHAR(50)
)
GO
INSERT INTO TableB VALUES(2, 'Priyanka', 'Female','IT')
INSERT INTO TableB VALUES(3, 'Preety', 'Female','HR')
GO
```

Example: Using EXCEPT Operator

The following query returns the rows from the left query that aren't present in the right query's results.

```
SELECT ID, Name, Gender, Department FROM TableA  
EXCEPT  
SELECT ID, Name, Gender, Department FROM TableB
```

Result:

ID	Name	Gender	Department
1	Pranaya	Male	IT

Example: Using NOT IN Operator

The same result can also be achieved using the NOT IN operator using the following query.

```
SELECT ID, Name, Gender, Department FROM TableA Where ID NOT IN (Select ID from TableB)
```

Result:

ID	Name	Gender	Department
1	Pranaya	Male	IT

So, what is the difference between EXCEPT and NOT IN operators in SQL Server?

The EXCEPT operator filters duplicate rows and return only DISTINCT rows from the left query that aren't in the right query's results, whereas NOT IN does not filter the duplicates rows. To understand this, insert the following row into TableA

```
INSERT INTO TableA VALUES(1, 'Pranaya', 'Male','IT')
```

Now execute the following EXCEPT query. Notice that we get only the DISTINCT rows in the result set.

```
SELECT ID, Name, Gender, Department FROM TableA  
EXCEPT  
SELECT ID, Name, Gender, Department FROM TableB
```

Result:

ID	Name	Gender	Department
1	Pranaya	Male	IT

Now execute the following query. Notice that the duplicate rows are not filtered in the result set.

```
SELECT ID, Name, Gender, Department FROM TableA Where ID NOT IN (Select ID from TableB)
```

Result:

ID	Name	Gender	Department
1	Pranaya	Male	IT
1	Pranaya	Male	IT

The EXCEPT operator expects the same number of columns in both the queries, whereas NOT IN, compares a single column from the outer query with a single column from the subquery. In the following example, the number of columns is different.

```
SELECT ID, Name, Gender, Department FROM TableA  
EXCEPT  
SELECT ID, Name, Gender FROM TableB
```

The above query would produce the following error.

Msg 205, Level 16, State 1, Line 1

All queries combined using a UNION, INTERSECT, or EXCEPT operator must have an equal number of expressions in their target lists.

The NOT IN operator compares a single column from the outer query with a single column from the subquery. In the following example, the subquery returns multiple columns

```
SELECT ID, Name, Gender, Department FROM TableA Where ID NOT IN (Select ID, Name from TableB)
```

The above query would produce the following error.

Msg 116, Level 16, State 1, Line 2

Only one expression can be specified in the select list when the subquery is not introduced with EXISTS.

In the next article, I am going to discuss the **INTERSECT Operator** in SQL Server with some examples. Here, in this article, I try to explain the EXCEPT Operator in SQL Server with Examples. I hope you enjoy this EXCEPT Operator in SQL Server with Examples article. I would like to have your feedback. Please post your feedback, question, or comments about this EXCEPT Operator in SQL Server with Examples article.